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ASSIGNMENT 1 WIND POWER PREDICTION – REPORT

Group name: Group B

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1 METHODOLOGY

1.1 DATA CLEANING

Two percent of the data was missing (Figure A.1) and was filled using three different methods, inspired by the guidelines. *Persistence fill* replaces a missing value with the last observation; it is simple, handles gradual changes well, but creates "flat spots" if many values are missing consecutively. *Moving average fill* takes the last m hours of data and inserts to the missing value. This handles outliers well, but can cause issues with smoothing out peaks and troughs. It is also heavily dependent on the parameter m. *Hourly climatology* takes the average of each hour and inserts this value into the corresponding hour of a missing value. It captures diurnal patterns but ignores current weather conditions.
Further, the u-v coordinates are converted to polar (speed, direction) for a more physical wind representation. Speed correlates to kinetic energy and direction captures the influence of turbine orientation or obstacles.

1.2 DATA EXPLORATION

Firstly, a correlation heatmap is generated to quantify relations between variables. From Figure A.2 a very positive correlation between wind speed and power can be identified, indicating that wind speed will be the primary predictor of the output power. As the correlation heatmap is not suitable for circular variables like the wind direction, no correlation to the power is shown. Additionally, near-unity correlation between 10m and 100m measurements for both wind speed and direction suggest data redundancy.
To further explore the relation between the wind speed and power, two scatter plots are presented in Figure A.3 showing the power curves for both heights. As expected, they share a similar "S"-shape. A "cut-in" speed can be estimated at which the power rises above 0, as well as a "rated" speed where the power reaches its maximum. Given that both curves are highly similar, only showing a minor shift, reinforces the suggestion to proceed with only one altitude.
The output power's dependence on wind direction, a second plot is presented in Figure A.4, isolating the average power curve across four wind direction quadrants. It can be seen, that for the wind going towards *NW* and towards *SW* the power curve is significantly lower than for the other directions. A likely explanation is that winds coming from east are blocked by some geographical obstruction - most likely other wind turbines. Consequently, this proves that power output is dependent on the wind direction, and this interaction must be

accounted for in the forecasting model.
Lastly, we explored the temporal patterns of wind power forecasting. Looking at the intra-day box plot in Figure A.5, only a small increase in power output at noon, compared to midnight, is present. While for the seasonal box plot in Figure A.6 are more distinct pattern can be seen, with the power output being the highest in the winter months.

1.3 Feature Engineering

Based on the insights from out data exploration, we designed new features to capture the data's patterns and improve the model's performance. To approximate the "S"- shape of the power curve, a categorical feature is created, separating the curve into three sections. Between $0-4\frac{m}{s}$ (cut-in), $4-20\frac{m}{s}$ (ramp-up) and $\geq 20\frac{m}{s}$ (max-power). As for the wind direction, we used the same separation as mentioned in subsection 1.2, creating a second categorical feature using the quadrants *NE*, *SE*, *SW* and *NW*. For the intra-day and seasonal behavior, cyclical time features are chosen to catch the temporal patterns.

1.4 Model Training

As shown previously, the relationship between wind speed and power output is nonlinear. Therefore, a Gradient Boosting decision tree model, XGBoost, and the Random Forest Regressor are chosen for training on the three preprocessed datasets: Persistence Fill, Rolling Average Fill, and Hourly Climatology Fill. The models were selected due to their ability to handle noisy nonlinear time-series data. These models are also compared to the Linear Regression model to further illustrate the nonlinearity of the data. The model's performance is then evaluated using cross-validation, and the results are compared using the Mean Absolute Error (MAE) and the R-squared score (Coefficient of Determination).
After cross-validation, it was found that all three models had relatively high R-squared scores and low MAE values. However, for all three datasets, it can be seen that XGBoost had the best results; therefore, this model is chosen for further hyperparameter optimization. A marginally higher accuracy can also be noted using the Persistence Fill dataset. Consequently, this method for filling in missing power output data provides the most accurate results.
Hyperparameter optimization is done using RandomizedSearchCV with the goal of maximizing the R-squared score. After optimization, the best R-squared score achieved on the training data was approximately 0.78. Afterwards, the test data (WeatherForecastInput A) is also fed into the same preprocessing pipeline as the training set and fit to the optimized XGBoost model. The prediction results are compared to the solution (Solution A) in the provided "forecast evaluation" notebook.

2 Results

The final R-squared score for the "A" test set is 0.86, which is noticeably higher than the training set results, and the MAE is 0.095. We propose that this is due to the absence of missing data in the test set and the dataset having fewer samples, which would result in less noise. Lastly, we plot the predicted power output and the "Solution A" power data to visually compare the results Figure A.7.
The same process of data preparation, model training, hyperparameter optimization, and testing was done during the meeting on the "B" data set. In this case, the optimized XGBoost model had an R-squared score of 0.78 on the training set and a final R-squared score of 0.81 on the test data.

3 Discussion

In general, the MAE was low, and the R-squared score was relatively high for the power output predictions on the test datasets for the optimized XGBoost model. Like most decision tree-based models, it struggles with extremes and outliers in data. A notable example of this is that the model never predicted a power output of 0 or 1 for either of the test datasets. Perhaps the model accuracy could be further improved through some feature engineering, for example, by increasing the importance of seasonal fluctuations when predicting for a specific month, or by implementing Model Output Statistics (MOS) features to account for discrepancies between the models predicted power output and the observed power output based on historical data. Adding more meteorological features could also help, such as temperature and pressure.

A Appendix

Additional Figures and Tables

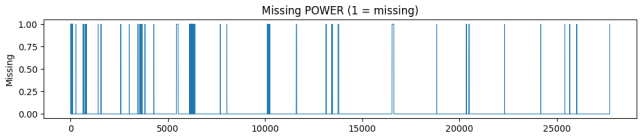


Figure A.1: Missing values in the training dataset

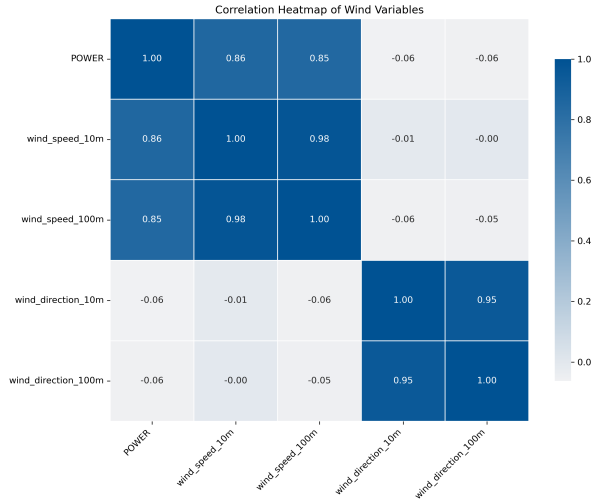


Figure A.2: Correlation Heatmap of all variables in the training data

References

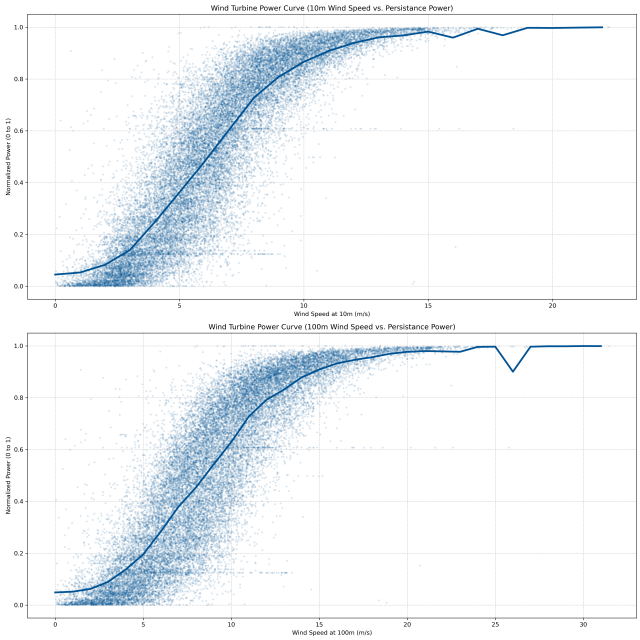


Figure A.3: Power curves for wind speeds at 10m and 100m

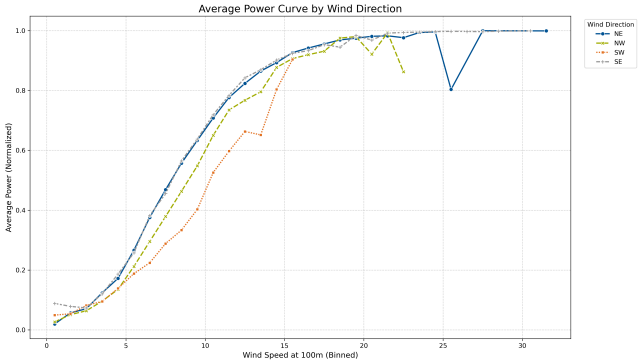


Figure A.4: Average power curve at four quadrants.

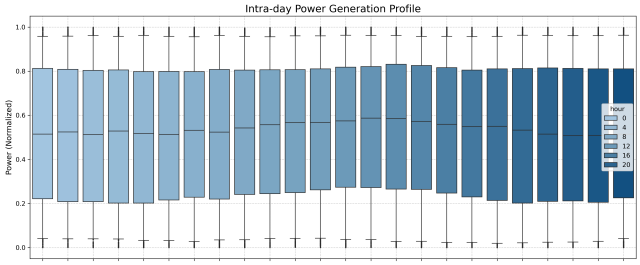


Figure A.5: Intra-day pattern of the power output.

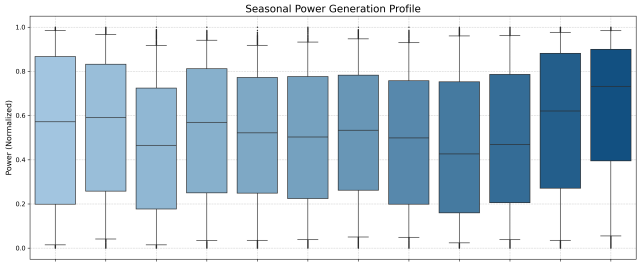


Figure A.6: Seasonal pattern of the power output.

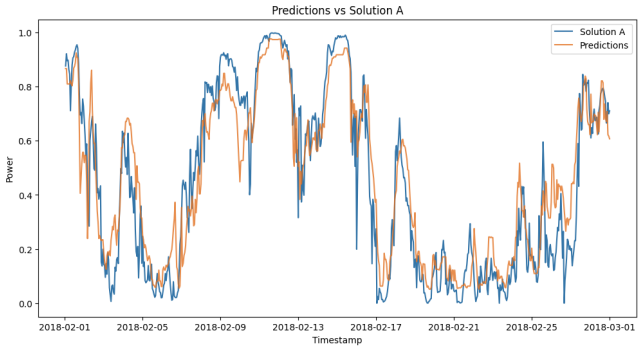


Figure A.7: Predicted Power Output vs Solution A Power data